

TRAFFIC IMPACT STUDY

1100 Peachtree Multifamily — 240 DU

Prepared May 14, 2026

PROJECT NO.	—
PREPARED BY	Sample Engineering, PE
REVIEWER	—
STUDY TYPE	Traffic Impact Study
SITE COORDINATES	33.7858, -84.3848

PE Stamp

Signature

Date

Traffic Impact Study

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10 INTERSECTIONS	4 LOS DROPS	2 AT LOS E/F	21.9s WORST Δ DELAY
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Project & Inputs

Project name	1100 Peachtree Multifamily — 240 DU
Address	1100 Peachtree St NE, Atlanta GA 30309
Coordinates	33.7858, -84.3848
Land use	221 Multifamily Housing (Mid-Rise)
Size	240 dwelling units
Opening year	2027
Study radius	0.75 mi
Weather	clear
Background growth	1.5%/yr × 1 yr
Pass-by applied	10%
Internal capture	0%

PM Peak Trip Generation

Daily trips	1271
AM peak trips	67
PM peak trips	101 (62 in / 39 out)

Trip Generation by Period

Period	Raw	Pass-by	Int. cap.	External	In	Out
AM Peak	75	8	0	67	23	44
PM Peak	113	12	0	101	62	39
Saturday Midday	124	13	0	111	56	55
Daily	1,271	0	0	1,271	636	635

Affected Intersections (10)

Intersection	Dist (mi)	Trips	Existing LOS	Future LOS	Δ delay (s)	Q95 (ft)
Peachtree St NE & 14th St NE	0.18	14	C	▲ D	8.3	320
Peachtree St NE & 10th St NE	0.34	9	C	C	5.3	155
Spring St NW & 14th St NW	0.51	6	D	▲ E	21.9	410
W Peachtree St NW & 17th St NW	0.42	5	B	▲ C	2.3	95
Juniper St NE & 11th St NE	0.29	7	B	B	1.8	78
Crescent Ave NE & 12th St NE	0.22	8	B	B	1.9	62
W Peachtree St NW & 14th St NW	0.31	4	F	F	14.5	580
Peachtree St NE & 17th St NE	0.46	6	C	C	2.6	138
Spring St NW & 17th St NW	0.63	3	C	C	2.4	162
Peachtree St NE & 12th St NE	0.12	11	C	▲ D	8.7	285

Recommended Mitigations

Peachtree St NE & 14th St NE [MODERATE]

Re-time NB through phase +5s and add EB right-turn overlap to absorb projected southbound peak inflow.

Spring St NW & 14th St NW [MAJOR]

Add eastbound right-turn lane (200 ft taper + 100 ft storage) and adjust signal split 8s toward EB.

W Peachtree St NW & 17th St NW [MINOR]

Minor signal-timing optimization, 3-5s green-shift toward critical southbound through phase.

W Peachtree St NW & 14th St NW [MAJOR]

Already operating at LOS F under existing conditions. Project does not meaningfully degrade further; significant geometric improvements would be required to recover LOS regardless of project.

Peachtree St NE & 12th St NE [MODERATE]

Re-time critical NB through phase +4s. Modest queue impact at 95th-percentile back-of-queue.

Monte-Carlo Sensitivity

Iterations	100
Mean worst Δ delay	7.20s
P10 / P50 / P90	5.40s / 7.20s / 9.10s
Probability ≥ 1 LOS drop	92%
Probability any LOS E/F	78%
Expected LOS drops	3.40

Findings

- Project will generate 1,271 new daily vehicle trips, with 101 during the PM peak hour (62 inbound / 39 outbound) after 10% pass-by capture per ITE TGM Appendix B.
- Existing volumes were grown by 1.50%/yr over 1 year ($\times 1.015$) to the 2027 opening-year horizon.
- 10 signalized intersections fall within the 0.75-mi study area; 4 are projected to drop at least one LOS grade after build-out.
- 2 intersections are projected to operate at LOS E or F under the build condition and require formal mitigation per City of Atlanta DOT TIS guidance.
- 1 intersection (W Peachtree St NW & 14th St NW) operates at LOS F under existing conditions; project contribution is marginal at this location.
- Monte-Carlo sensitivity (100 runs, $\pm 10\%$ trip-rate / $\pm 15\%$ existing-volume): worst-case delay change is 7.2s (median); 80% range 5.4–9.1s. Probability of ≥ 1 LOS drop: 92% (high confidence the LOS impacts shown are robust to input uncertainty).

Methodology

- Trip generation uses ITE Trip Generation Manual 11th-Edition average rates for the selected land-use code, computed for AM peak, PM peak, Saturday midday, and daily totals. Saturday-midday rates are estimated as a published industry multiple of the PM peak rate by land-use category.
- Pass-by and internal-capture credits are applied at the PM peak per ITE's Pass-By Trip Generation Manual (3rd Edition) and ULI Mixed-Use Internal Capture defaults; only the residual external trips are assigned to off-site intersections.
- Existing intersection volumes are grown to the opening-year horizon at the user-supplied annual growth rate (default 1.5%/yr) before the capacity analysis.
- Weather adjustment follows HCM 6th-Edition Ch. 11 (rain/snow capacity reduction): clear 1.00, light rain 0.95, heavy rain 0.86, light snow 0.86, heavy snow 0.70. The factor multiplies the saturation flow at every intersection.
- Off-site impact is screened for all signalized intersections within the study radius. New trips are assigned by inverse-distance weighting (clamped at 100m), normalised to sum to 100% of the period's external trip total.
- Intersection-level control delay uses the HCM signalized-intersection model $d = d_1 + d_2$ (Webster uniform delay + Akçelik/HCM incremental-delay term) with a 90s cycle, $g/C = 0.45$, 1,800 vphpl saturation flow ($\times$ weather factor), 15-minute peak analysis period ($T = 0.25$ hr) and pretimed-signal incremental-delay factor $k = 0.5$.
- Approach-level analysis splits each signal's inflow across NB/SB/EB/WB approaches (deterministic per-signal

allocation perturbed $\pm 15\%$ from a 30/25/25/20 base) and assigns added trips to each approach by cosine-similarity to the bearing of the project relative to the signal. Per-approach v/c, control delay, LOS, and 95th-percentile back-of-queue length (HCM Eq. 19-50, $Q_{95} \approx Q_1 \times 1.65 \times 25 \text{ ft/veh}$) are reported.

- Level of Service is assigned from HCM 6th-Edition signalized-intersection control-delay thresholds (Exhibit 19-8): A $\leq 10s$, B $\leq 20s$, C $\leq 35s$, D $\leq 55s$, E $\leq 80s$, F $> 80s$.
- Monte-Carlo sensitivity perturbs the project trip rate by $N(1, 0.10)$ and the baseline existing volume by $N(1, 0.15)$ over 100 iterations and reports the resulting distribution of worst-case delay change and probability of any LOS drop.
- Mitigations are screening-level recommendations sized to the projected delay change, not full Synchro/SimTraffic optimization runs. A formal TIS submittal should validate these recommendations with detailed traffic counts and signal-timing analysis.

Citations & Methodology

- ITE Trip Generation Manual, 11th Edition — peak and daily rates by land-use code.
- ITE Pass-By Trip Generation Manual, 3rd Edition — pass-by capture defaults.
- HCM 6th Edition, Chapter 11 (Capacity and LOS) — saturation flow + weather adjustment.
- HCM 6th Edition, Chapter 19 (Signalized Intersections) — Eq. 19-13 control delay, Exhibit 19-8 LOS thresholds.
- HCM 6th Edition, Chapter 31 — 95th-percentile back-of-queue methodology.
- ULI Mixed-Use Development Internal Capture defaults.
- City of Atlanta DOT — Traffic Impact Study Guidelines.
- Sample report generated by Simple Impact Studies — simpleimpactstudies.com

